

RESEARCH ARTICLE

Energy yield of all thin-film perovskite/CIGS tandem solar modules

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Abstract

Perovskite/CuIn_{1-x}Ga_xSe₂ (CIGS) tandem photovoltaics (PV) promises high power conversion efficiencies in combination with the advantages of a light weight and an all thin-film PV technology. However, the complexity of perovskite/CIGS tandem solar module architectures requires careful optimization of the layer stack under realistic solar irradiation conditions. Here, we provide a systematic numerical study on the energy yield (EY) of perovskite/CIGS tandem solar modules, optimizing the device architecture with regard to irradiance in various climate zones. In particular, variations of the spectral irradiation and the average photon energy are of decisive importance for the location-specific optimization of the device architecture. Compared with the reference single-junction CIGS thin-film PV technology, we demonstrate a strong improvement in EY (>30%) in perovskite/CIGS thin-film PV for perovskites of a wide range of bandgaps (1.55–2.0 eV), reaching up to 52% improvement in EY for the optimal bandgap (around 1.8 eV). Of the two most favored architectures, the two-terminal and four-terminal devices, perovskite/CIGS tandem solar modules with low bandgap (~1.55 eV) perovskite absorbers in the four-terminal architecture outperform those in the two-terminal architecture in average by 3.5% relative. However, for wide perovskite bandgaps ranging from 1.75 eV to 1.85 eV, both architectures perform comparably. The improvements in EY for perovskite/CIGS tandem solar modules highlight the potential of this technology but also the vital need for light management in tandem photovoltaics.

KEYWORDS

energy yield modelling, perovskite/CIGS tandem photovoltaics, realistic conditions

1 | INTRODUCTION

Novel photovoltaic (PV) materials and concepts are of key importance to uphold the ongoing increase of record power conversion efficiencies (PCEs) of PV.^{1,2} One strategy to realize high PCEs are tandem solar cells, which reduce the charge carrier thermalization losses by harvesting high energy photons of the sun at a high voltage in the top solar cell and low energy photons in the low bandgap bottom solar cell (see Figure 1A). According to the detailed balance theory, the limit in PCE of tandem solar cells is approximately 46% and, thereby, much higher than the limit of single-junction (SJ) solar cells of approximately 33%.^{3,4} With the recent rise of perovskite PV, for the first time a low-cost, high efficiency, and wide bandgap absorber material is available that can be combined with the market-dominating PV tech-

nologies based on crystalline silicon ($E_g \sim 1.1$ eV) or thin-film copper indium gallium diselenide (CIGS; $E_g \sim 1.2$ eV) to realize a high efficiency tandem solar module.^{5,6} In 2015, the first perovskite/CIGS tandem solar cell was reported, reaching a remarkable PCE of up to 20.5%.⁷ Further optimizations led to a fast and substantial increase of the PCEs up to 23.9% for mechanically stacked tandem architectures^{8,9} and 22.4% for monolithic tandem architectures¹⁰ on the cell level. Moreover, the first perovskite/CIGS tandem modules have been prototyped, reaching PCEs of up to 17.8%.¹¹ Next, to the high PCE, the technology offers the ease of an integrated all thin-film technology, which can be fabricated on thin, lightweight, and flexible substrates.

Despite the fast advancement of the technology, further progress in PCE of perovskite/CIGS tandem solar modules has to be made such that the higher PCE of this technology justifies the higher complexity

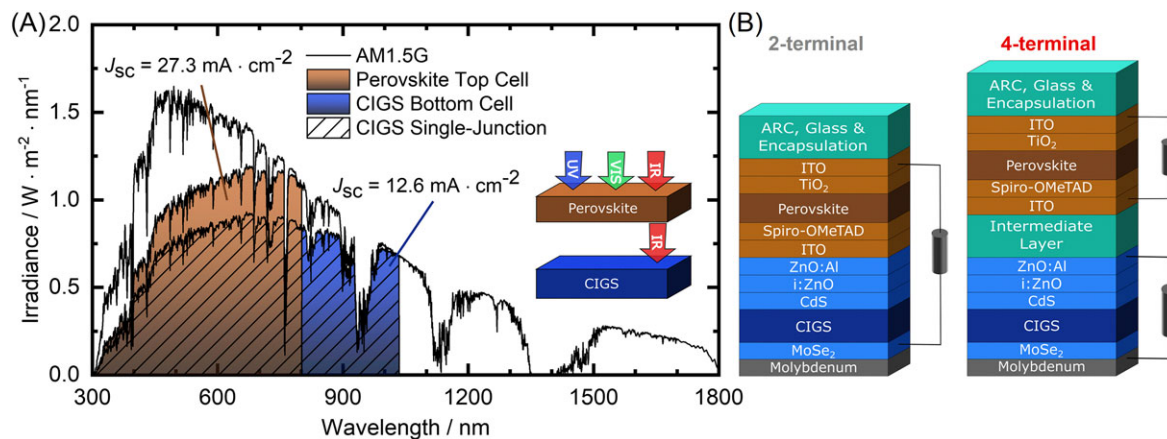


FIGURE 1 A, Illustration of the utilizable power density of a CIGS SJ and a perovskite/CIGS tandem solar module. Compared with the CIGS SJ solar module (striped region), lower thermalization losses in the tandem solar module (brown and blue region) lead to a remarkably increased utilizable power density. The maximum short-circuit current density (J_{sc}) of both sub cells are indicated under the assumption of perfect carrier collection efficiency. B, two-terminal and four-terminal perovskite/CIGS solar module architectures with a schematic layout of their electrical connection are outlined [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jpp.3091)]

compared with the SJ record PCEs of CIGS solar cells (22.9%)¹² and perovskite solar cells (23.3%).¹³ As a consequence, perovskite/CIGS tandem solar modules require optimization in order to reduce optical losses and maximize the power generation output.^{14–16} To date, two types of architecture are researched and developed: A, mechanical stacking of perovskite top solar cell and low-bandgap bottom solar cell without direct electrical interconnection, called “four-terminal architecture” (4T); and B, monolithic connection of the top and bottom solar cells in series, called “two-terminal architecture” (2T). The different architectures, illustrated in Figure 1B, involve distinct advantages and disadvantages. In the 4T architecture, the top and bottom solar cell can be operated at their individual maximum power point (MPP), but significant reflection and absorption losses are induced because of the requirement of thicker transparent electrodes for lateral charge transport.¹⁶ In contrast, the 2T architecture exhibits thinner transparent electrodes and, in turn, lower optical losses, but the monolithic interconnection of the top and bottom solar cell results in losses because of discrepancies in the currents of the subcells for variable irradiance spectra. Therefore, careful optimization of the annual energy yield (EY) under realistic solar irradiation conditions is required, in order to identify the realistic potential of perovskite/CIGS tandem solar modules and to elucidate the superiority of one architecture over the other. In the context of perovskite-on-silicon tandem solar cells, EY modelling is already used to perform optimizations of device architecture and to model the performance limitations.^{17–21}

This work provides a systematic study on the EY of latitude-tilted (max. 45°) perovskite/CIGS tandem solar modules for various locations in the United States of America. We employ optical simulations, combining the transfer-matrix method (TMM) and Beer-Lambert's law, for the accurate optical modelling of the perovskite/CIGS solar module device architecture. Combining these optical simulations with basic electrical simulations and detailed irradiance data, allows studying the EY of perovskite/CIGS solar modules in the 2T as well as 4T architecture for a typical meteorological year at exemplary locations in various climate zones. At this point, it shall be noted that we disregard temperature variations in this study for now as it focusses on

the optical modelling of perovskite/CIGS tandem PV. Furthermore, we quantify the major loss mechanisms occurring in perovskite/CIGS solar modules, such as reflection, absorption, and current matching losses. Additionally, we investigate the impact of variations in spectral irradiance, in particular the average photon energy (APE), on the EY and optimum device architecture of perovskite/CIGS tandem solar modules. For these investigations, we choose methyl ammonium lead iodide ($\text{CH}_3\text{NH}_3\text{PbI}_3$) as the starting perovskite absorber material, since it is commonly used as a reference material and displays similar optical properties as the highest performing multi-cation perovskite absorbers. Nevertheless, the bandgap of metal-halide perovskites can be tuned arbitrarily in the range of 1.55 eV to 2.3 eV by compositional variations of the halide in the perovskite crystal structure according to $\text{CH}_3\text{NH}_3\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$.²² Therefore, we study the EY of perovskite/CIGS tandem solar modules in 2T and 4T architecture for artificial perovskite absorber materials of bandgaps ranging from 1.55 eV to 2.0 eV. It shall be noted that we disregard the complex key issue of long-term stability of perovskite PV at this stage, as this is beyond the scope of this paper which focuses on the optical aspects of perovskite/CIGS solar modules.

2 | METHOD: ENERGY YIELD MODELLING

The flow of the EY modelling platform used in this study to determine the annual EY for the investigated architectures of perovskite/CIGS tandem solar modules is schematically depicted in Figure 2A. A more detailed description can also be found in a previous study.²¹ In general, the EY modelling consists of four software modules that address the calculation of A, the irradiance based on meteorological data, B, the absorptance based on the device architecture, C, the electrical behavior based on electrical characteristics of the device, and D, the annual EY.

The calculation of the irradiance relies on hourly and spectrally resolved typical meteorological year (TMY3) data and is available for a set of locations in the United States of America.²³ Initially, clear sky irradiance with a specular and diffuse fraction is determined using

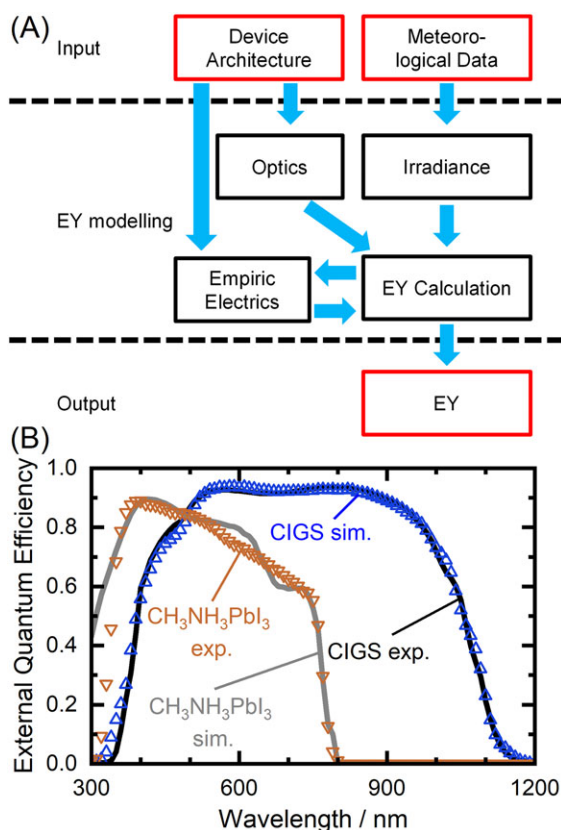


FIGURE 2 A, Schematic flow of the energy yield modelling platform. B, The measured and calculated external quantum efficiency (EQE) of each subcell of the perovskite/CIGS tandem solar module obtained from the optical model. The optical models are optimized with regard to measured EQEs of reference state-of-the-art record solar cells without module encapsulation^{24,25} [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the simple model of the atmospheric radiative transfer of sunshine (SMARTS),²⁴ where the diffuse fraction is assumed to exhibit a Lambertian distribution over the whole sky. Subsequently, a simple model for the cloud coverage is applied that assumes irradiance originating from clouds to have the same spectral distribution as specular clear sky irradiance.

The optical response of the investigated architectures is simulated using TMM for thin films that require a treatment of coherent forward and backward propagating electromagnetic waves. If “optically-thick” layers are apparent in the layer stack, a series expansion of the Beer-Lambert’s law is sufficient to account for multiple reflections at the interfaces to adjacent layers. The optical calculation makes use of complex refractive indices and layer thicknesses of the reference architectures.^{25,26} For calculations with wide bandgap perovskite absorbers, optical data of mixed-halide perovskites available in a previous study²² is interpolated to obtain complex refractive indices with a bandgap resolution of 0.05 eV (see Figure S1).

The architecture of the investigated tandem solar modules (Figure 1B) comprises of a combination of a perovskite top solar cell, a CIGS bottom solar cell as well as module specific layers, ie, an antireflection coating (ARC), top cover glass, and an encapsulation material that we assume to be ethylene vinyl acetate (EVA). The perovskite solar cell consists of an indium tin oxide (ITO) front electrode, a titanium

dioxide (TiO₂) electron transport layer, the perovskite absorber itself, a 2,2,7,7-tetrakis(N,N-di-4-methoxyphenylamino)-9,9-spirobifluorene (Spiro-OMeTAD) hole transport layer, and a thin and thick transparent ITO back electrode in the 2T and 4T architecture, respectively. Even though the low thermal stability of Spiro-OMeTAD hinders perovskite annealing in the 2T architecture,²⁷ we consider it as the hole transport material in the 2T architecture to maintain comparability between both architectures. Nevertheless, the choice of the hole transport material only has small impact on the optical performance, as can be seen from comparative simulations with 30 nm of nickel oxide (NiO) as hole transport material (see Figure S3). The selected perovskite absorber material is CH₃NH₃PbI₃ with a bandgap of 1.55 eV, since CH₃NH₃PbI₃ is a commonly used reference material in the field. Its optical properties are very similar to those of the recent most stable and highest performance perovskite solar cells employ multi-cation perovskites.²⁸ Additionally, we consider the transition between the perovskite absorber and the Spiro-OMeTAD as a mixed material consisting of two equal fractions of both materials. Thus, this mixed layer accounts for lower reflectivity because of the roughness of this interface. The improved match of the external quantum efficiency (EQE) of this approach with measurement results was already demonstrated in a previous study.²⁶ The front electrode of the CIGS cell consists of aluminum-doped zinc oxide (ZnO:Al) and intrinsic zinc oxide (i-ZnO). Furthermore, the CIGS cell involves a cadmium sulfide (CdS) buffer layer, the CIGS absorber layer, a molybdenum diselenide (MoSe₂) layer, and a molybdenum (Mo) back contact. It shall be noted that we disregard shadowing by electrical connections for all tandem architectures as well as the SJ references. We assume the architecture of the CIGS cell to be as in a previous study,²⁵ except disregarding the magnesium fluoride ARC. The 4T architecture requires an additional intermediate insulating layer to separate both cells electrically. In earlier studies, air was commonly used as an intermediate layer.^{6,29} However, highly transparent polymers or silicones with higher refractive indices around 1.5, like commonly used EVA, should aid in reducing reflection losses because they enable a smoother transition of refractive indices between the top and bottom solar cell.^{30,31} Therefore, we consider EVA as the intermediate layer in our simulations. We calibrate the optical models of both sub cells with regard to the EQE of the reference architectures of CIGS and opaque CH₃NH₃PbI₃ solar cells (as given in Figure 2B) by adjusting the layer thicknesses of all layers and the collection efficiencies of the absorbing layers. All layer thicknesses for both architectures used in this work are listed in Figure S2 and the sources of optical data are provided in Table S1.

Combining, finally, the orientation of the tandem solar module, the course of the sun, the incident irradiance, the collection efficiency, and the layer-resolved absorption (see Figure S3), we calculate the generated photocurrent J_{sc} . Subsequently, we determine the MPP of the current-density voltage (J - V) characteristic with the use of the single-diode equation. Thus, we obtain the power output for every hour of the year. With regard to the electrical properties of various wide bandgap perovskite materials, we assume a comparable electrical performance of the perovskite top solar cells by maintaining a fill factor (FF) of 80% under an illumination of air-mass 1.5 global (AM1.5g) solar irradiation for all bandgaps (see Table S2). Furthermore, for each bandgap, we calculate the dark saturation current density J_0 based on

the black-body radiation at a temperature of 300 K and the assumption of a perfect EQE.¹⁹ This approach results in a linear increase of the open-circuit voltage (V_{OC}) with increasing bandgap (see Figure S4), which is in agreement with similar studies.¹⁹

Based on the above-outlined method, we carry out optimizations of the device architecture of perovskite/CIGS tandem solar modules in 2T and 4T architecture with the objective to quantify reflection, parasitic absorption, and current matching losses. Furthermore, we determine the dependency of the perovskite absorber thickness and the perovskite bandgap on variations in spectral irradiance. The selection of locations intends to cover various classes of the Köppen climate classification.³² A list of all considered locations and their corresponding Köppen climate classification is provided in Table S2. Therefore, we perform EY calculations for a set of locations in the United States of America, covering different climate zones.

3 | RESULTS

First, we compare the overall EY of latitude-tilted perovskite/CIGS tandem solar modules in 2T and 4T architecture with the reference CIGS SJ solar module for various locations in the United States of America, covering a broad range of climate zones (see Figure 3A). For each location, the 2T architecture requires an optimization of the perovskite absorber thickness such that losses arising from mismatch in the current density of the perovskite top solar cell and the CIGS bottom solar cell are minimized. It should be kept in mind that the choice of the charge transport material has a small impact on the

optical performance layers in all thin-film perovskite/CIGS tandem PV (see Figure S3). Consequently, to minimize current matching losses in the 2T architecture, the perovskite layer thickness should be optimized for each layer stack individually.

For all considered locations, the perovskite/CIGS tandem solar modules significantly outperform the reference CIGS SJ solar module in terms of EY by 31%–36% and 35%–41% for the 2T and 4T architectures, respectively (see Figure 3B). Also, in terms of PCE in Figure S5, determined under standard test conditions (STC), we find a relative increase of approximately 32% for the 2T architecture (absolute PCE = 25.7%, PCE_{top} = 16.1%, PCE_{bottom} = 9.6%) and a relative increase of approximately 30% for the 4T architecture (absolute PCE = 25.3%, PCE_{top} = 17.8%, PCE_{bottom} = 7.5%) compared with reference CIGS SJ solar modules (absolute PCE = 19.5%). Thereby, our EY study demonstrates the promise of perovskite/CIGS tandem PV as a future high performance and all thin-film PV technology. Moreover, we identify a relative increase in PCE of 35% for the 2T architecture and of 32% for the 4T architecture compared with an opaque $CH_3NH_3PbI_3$ SJ solar module (absolute PCE = 19.1%). Despite having a lower absolute PCE, the opaque $CH_3NH_3PbI_3$ SJ solar module exhibits a higher EY in all investigated locations by 6.9% in average compared with the reference CIGS SJ solar module. Furthermore, the EY of the 4T architecture for a tandem module based on $CH_3NH_3PbI_3$ is higher than the EY of the 2T architecture (around 3.5% relative) for all locations, whereas the PCE shows the reverse trend. Consequently, the 4T architecture shows a higher specific yield of 2226 kWh/(kWp·a) than the 2T architecture with a specific yield of 2123 kWh/(kWp·a) for Phoenix, Arizona. This

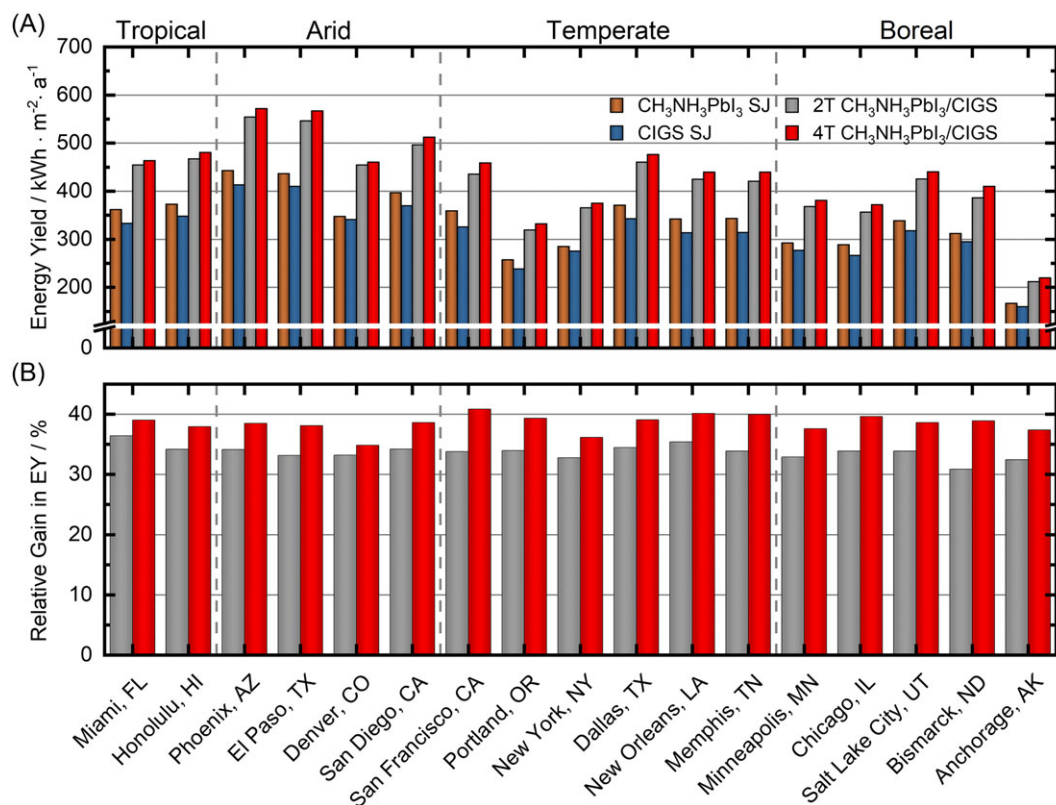


FIGURE 3 A, Energy yield (EY) of an opaque $CH_3NH_3PbI_3$ SJ (brown), CIGS SJ (blue), and $CH_3NH_3PbI_3/CIGS$ tandem solar modules in 2T architecture (grey) and 4T architecture (red) for various locations in the United States of America. The locations are ordered by the first order of their Köppen climate classification³¹ (Tropical, Arid, Temperate, Boreal). B, Relative gain in energy yield compared to the reference CIGS SJ solar module [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

emphasizes the importance of developing perovskite/CIGS tandem architectures with regard to realistic irradiation conditions rather than standardized test conditions.

Comparing the EY of perovskite/CIGS tandem solar modules at various locations, the EY in hot and sunny regions (Köppen climate classification: Tropical and Arid) in the southwest (Phoenix, Arizona) is higher than in cold and damp climatic regions (Köppen climate classification: Temperate and Boreal) in the north (Portland, Oregon) according to expectations. Moreover, the EY of the perovskite/CIGS tandem solar modules in coastal locations, for example San Diego, California, is slightly lower than in more inland locations (Phoenix, Arizona) because of the more humid and cloudier climate of the former. In arid climates with a large share of specular irradiation, tandem solar cells benefit strongly from tracking methods, one-axis by around 30% relative and two-axis by around 40% relative. In cloudy climate regions with a large share of diffuse irradiation, one-axis by around 26% relative and two-axis by around 32% relative (see Figure S6). It is worth noting that the relative increase of EY of perovskite/CIGS tandem solar modules compared with CIGS SJ solar modules correlates roughly with the APE of the average irradiance spectrum of each location (see Figure 3B and Table S2). We attribute this correlation to higher thermalization losses of the CIGS SJ solar module in locations with higher APE, which implies a larger contribution of photons of short wavelengths to the irradiance spectra.

The realistic irradiation conditions considered in the EY modelling differ from the STC by variations of: A, the solar irradiation spectra; B, the angle of incidence of the specular solar irradiation; and C, the contribution of diffuse solar irradiation. In order to discriminate the impact of these variations on the EY of perovskite/CIGS tandem solar modules, we derive three artificial irradiation scenarios from the realistic irradiation conditions in Phoenix, Arizona. The considered irradiation scenarios are: A, *Phoenix Real Spectra*, representing the real irradiance data; B, *Phoenix Invariant Spectra*, disregarding spectral variations by employing the AM1.5g spectrum normalized to the real irradiance intensities; C, *Phoenix all I_{specular}* , assuming all solar irradiation is specular; and D, *Phoenix all I_{diffuse}* , assuming all solar irradiation is diffuse (see Figure 4). Comparing the EY of perovskite/CIGS tandem solar modules for these solar irradiation scenarios, it is apparent that the relative differences in EY between the 2T architecture and 4T architecture for *Phoenix all I_{specular}* and *Phoenix all I_{diffuse}* merely vary from the reference *Phoenix Real Spectra*. In contrast, the relative difference in EY between the 2T and 4T architecture is lower and even reverses in sign for *Phoenix Invariant Spectra*, which agrees also with the trend of higher PCE under STC for the 2T architecture compared with the 4T architecture. Thereby, we demonstrate that variations of irradiation spectra in the reference *Phoenix Real Spectra* are the major cause of the lower EY enhancement for the 2T architecture compared with the 4T architecture. We can further conclude that the variations in the angle of incidence as well as the ratio of specular and diffuse irradiation are both of much lower relevance. From a device perspective, these findings imply that the variations of the spectra in the realistic irradiation scenarios impact significantly on the current matching in the 2T architecture. We attribute the lower impact of variations in the angle of incidence to the refraction of incident light into the encapsulation layer (typical refractive index of 1.45–1.5) in

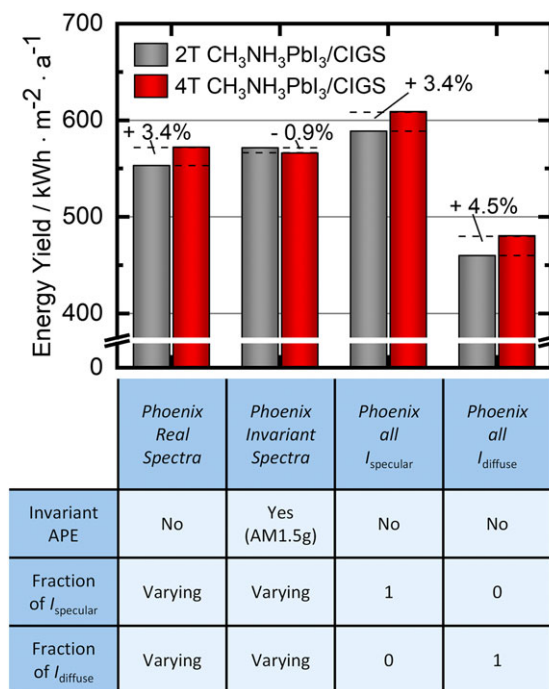


FIGURE 4 The energy yield of $\text{CH}_3\text{NH}_3\text{PbI}_3/\text{CIGS}$ tandem solar modules in 2T (grey) and 4T (red) architecture for various irradiation scenarios is shown. The characteristics of all investigated scenarios are summarized in the lower table. The corresponding irradiance spectra are depicted in Figure S7 [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jpp.3091)]

the solar module configurations, which reduces the conical solid angle of incident light acceptance on the solar cell to around 40° to 45° .

Having demonstrated the strong impact of the irradiation spectra on the EY of the perovskite/CIGS tandem solar modules in 2T architecture, we investigate the implications on the optimization of the perovskite absorber thickness next. In Figure 5A, we show the interdependence of the EY of the 2T architecture and the perovskite absorber thickness for the scenarios *Phoenix Real Spectra* and *Phoenix Invariant Spectra*. It is evident that the EY is highest close to the point of matching average current densities at MPP (J_{mpp}) of top and bottom cell for both scenarios. Here, J_{mpp} is the current density at MPP J_{mpp} averaged over a whole year. Moreover, the optimum perovskite absorber thickness for *Phoenix Real Spectra* ($d = 155$ nm, APE = 1.51 eV) is significantly smaller than for *Phoenix Invariant Spectra* ($d = 190$ nm, APE = 1.45 eV) and correlates with the decrease in APE for these spectra. In fact, this trend is apparent for the entire range of irradiation spectra of various locations distributed over the USA as shown in Figure 5B. Independent of the climatic conditions, the optimum perovskite absorber thickness decreases with increasing APE of the irradiation spectra for the considered location. Since a high APE implies a higher photon flux that can be harvested in the perovskite top solar cell, the optimum perovskite absorber thickness is reduced to maintain current matching.

It is worth stating that the scenario *Phoenix Invariant Spectra* also features a higher maximum in EY compared with *Phoenix Real Spectra*. We explain this by the fact that in this scenario, the perovskite absorber thickness is simply optimized with regard to a single spectrum (AM1.5g), which implies close to optimal current matching for the

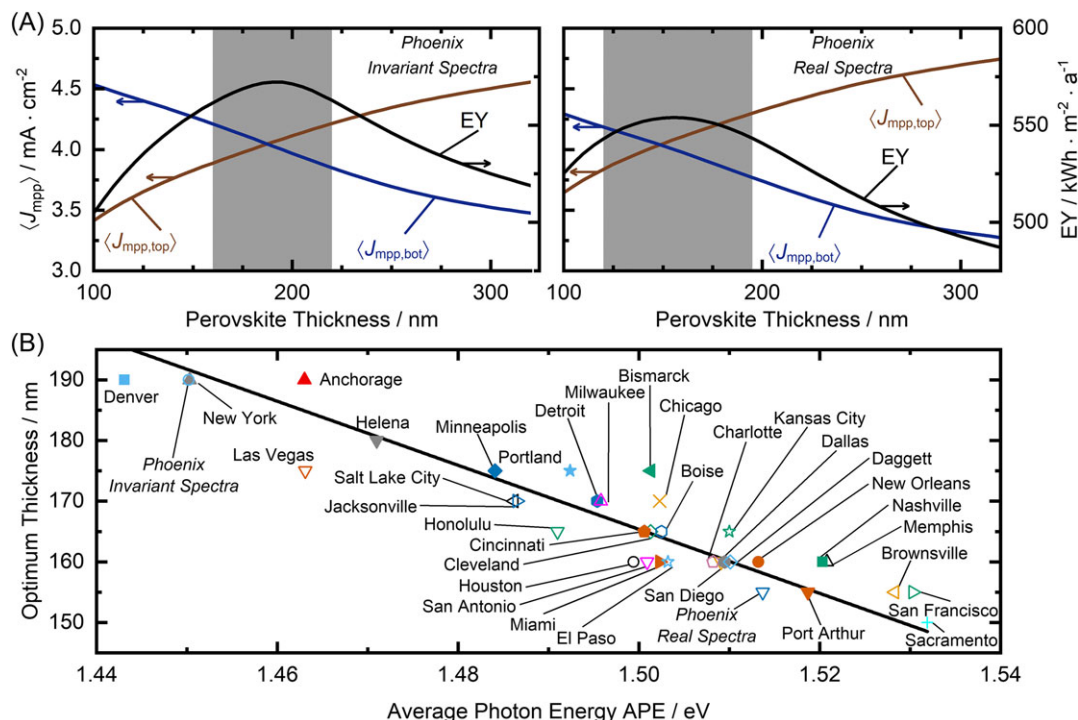


FIGURE 5 Average current densities at maximum power point (J_{mpp}) as a function of the perovskite absorber thickness d and the corresponding energy yield of the perovskite/CIGS tandem solar module for the irradiation scenario A, *Phoenix Real Spectra* and *Phoenix Invariant Spectra*. Perovskite absorber thicknesses that yield at least 98% of maximum EY are highlighted by grey areas. B, The optimum perovskite absorber thickness determined for all investigated locations is shown in dependence of the location's APE. An overview of all investigated locations is given in Table S2 [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

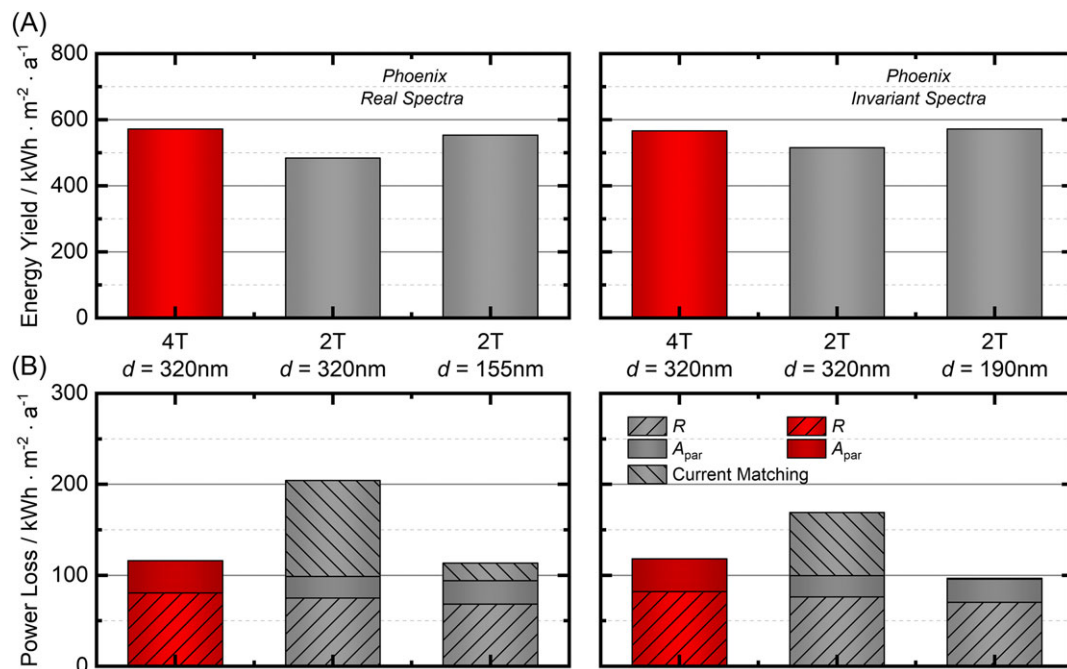


FIGURE 6 A, The annual EY for latitude-tilted CH₃NH₃PbI₃/CIGS tandem solar modules for the irradiance scenarios *Phoenix Real Spectra* and *Phoenix Invariant Spectra*. B, The energy losses attributed to the three major loss mechanisms, current matching, reflection R and parasitic absorption A_{par} , are indicated for each architecture respectively [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

entire year. Moreover, the improved current matching for *Phoenix Invariant Spectra* leads to a more pronounced maximum in EY compared with *Phoenix Real Spectra*, which is highlighted by the grey areas indicating perovskite absorber thicknesses that yield at least 98% of the maximum EY.

It shall be noted that adapting the architecture by varying the perovskite absorber thicknesses impacts also other key optical loss mechanisms. Therefore, in the next step, we quantify the overall EY (see Figure 6A) and reflection, parasitic absorption, and we discriminate current matching losses by turning them off subsequently (see

Figure S8) for three different architectures (see Figure 6B): A, the 4T architecture with the reference perovskite top solar cell ($d = 320$ nm); B, the 2T architecture with the same $d = 320$ nm; and C, the 2T architecture with an individually optimized perovskite absorber thickness d for *Phoenix Real Spectra* and *Phoenix Invariant Spectra*.

- Current matching: Comparing reflection, parasitic absorption, and current matching losses in detail (see Figure 6B) for A and B, it becomes apparent that the loss in EY associated to imperfect current matching in the 2T architecture is by far the most dominant loss mechanism. However, these current matching losses can be reduced significantly by 81% via an optimization of the device architecture (C), such that the EY of the optimized 2T architecture is comparable with the EY of the 4T architecture. Contrary to *Phoenix Invariant Spectra*, for *Phoenix Real Spectra*, even for an optimized perovskite absorber thickness, the intrinsic losses associated to current matching cannot be eliminated though.
- Reflection: Next to the current matching losses, the reflection losses impact the EY of both the 2T architecture and 4T architecture. The 4T architecture shows slightly higher reflection losses because

of the stark change in refractive index at the interfaces of the additional insulating layer in between the top and the bottom solar cell. In principal, the reflection losses for both architectures can be significantly reduced further by advanced light management concepts.^{33,34}

- Parasitic absorption: Another advantage of the 2T architecture over the 4T architecture is the lack of interlayers and lower thickness of transparent electrodes. Since these layers induce parasitic absorption, the EY of the 4T architecture is affected stronger (see Figure 6B). Furthermore, parasitic absorption losses are likely much more complex to reduce compared to reflection losses, as entirely novel contacting schemes would need to be developed.^{31,35}

According to the detailed balance limit of the PCE of tandem solar cells, the optimum bandgap of the perovskite top absorber is around 1.7 eV to 1.8 eV for the reference CIGS bottom solar cell ($E_g \sim 1.2$ eV).⁴ Consequently, as the next step, we expand the EY study on perovskite/CIGS tandem solar modules beyond $\text{CH}_3\text{NH}_3\text{PbI}_3$ ($E_g \sim 1.55$ eV) to wide bandgap perovskites with bandgaps of up to 2.0 eV (see

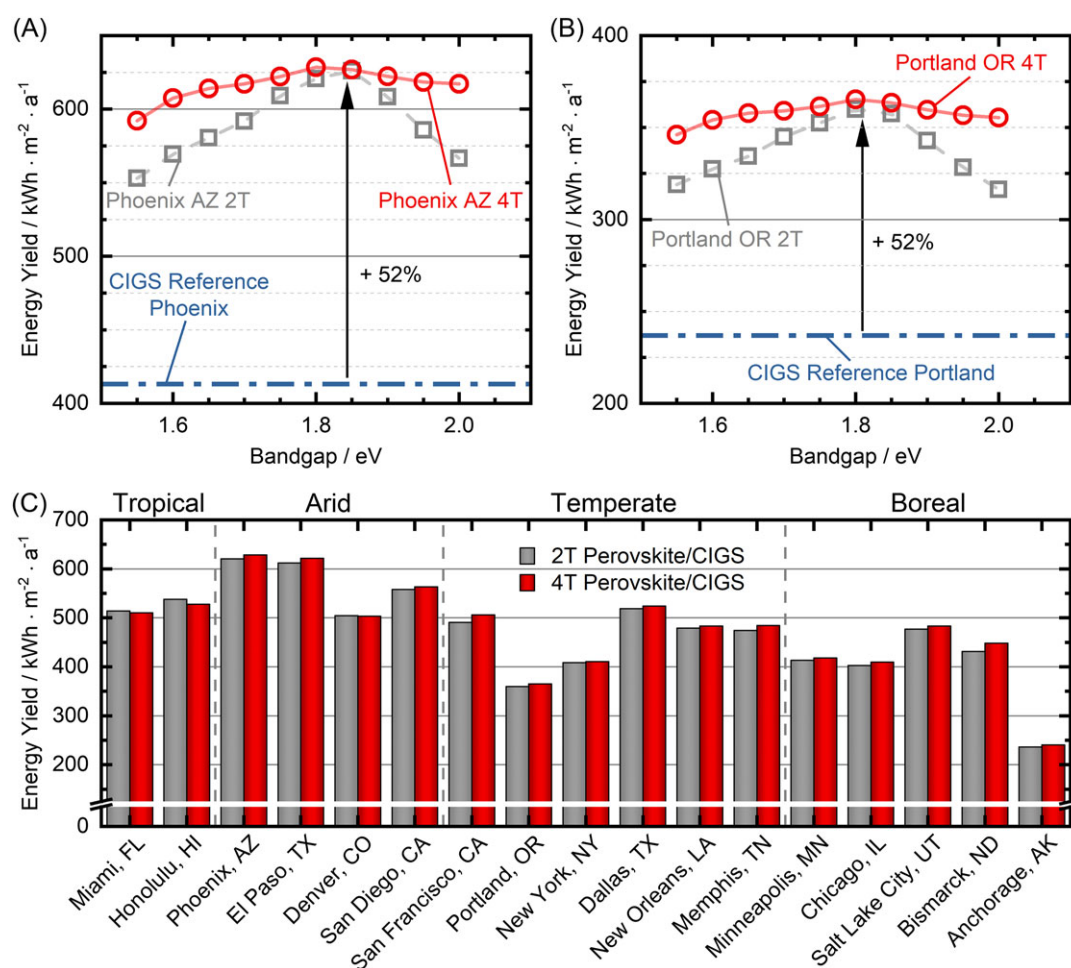


FIGURE 7 The energy yield for 2T (dashed lines) and 4T (solid lines) perovskite/CIGS tandem solar modules with different bandgaps of the perovskite ($\text{CH}_3\text{NH}_3\text{Pb}[\text{I}_{1-x}\text{Br}_x]_3$) top absorber is depicted for locations of two different climate zones, A, dry and sunny (Köppen climate classification: Arid) Phoenix, Arizona (black) and B, humid and cloudy (Köppen climate classification: Temperate) Portland, Oregon (red). The layer thickness of the perovskite absorber is optimized for the 2T architecture with a limit of 1000 nm; whereas in the 4T architecture, the perovskite absorber thickness was kept at that limit. A comparison at more common layer thickness of maximum 500 nm is provided in Figure S9. C. Overview of EY of wide bandgap (1.8 eV) perovskite/CIGS tandem solar modules in various locations in the USA. The locations are ordered by the first order of their Köppen climate classification³¹ (Tropical, Arid, Temperate, Boreal) [Colour figure can be viewed at wileyonlinelibrary.com]

Figure S1). The considered generic locations are Phoenix, Arizona, and Portland, Oregon (see Figure 7). For the 2T architectures, we optimize the perovskite absorber thickness again, but limited to a maximum of 1000 nm considering typical state-of-the-art perovskite absorber layers. To ensure comparable amounts of thermalization losses, we keep the perovskite absorber thickness in the 4T architecture at the maximum of 1000 nm.

Overall, the EY of perovskite/CIGS tandem solar modules with wide bandgap perovskite increases by up to 52% compared with the reference CIGS SJ solar module. For the 2T architecture, this is a further relative increase by 13% compared with $\text{CH}_3\text{NH}_3\text{PbI}_3$ /CIGS tandem solar modules (E_g of $\text{CH}_3\text{NH}_3\text{PbI}_3 \sim 1.55$ eV). In contrast, the 4T architecture only benefits by 6% relative from the use of a wide bandgap perovskite absorber. Contrary to $\text{CH}_3\text{NH}_3\text{PbI}_3$ absorbers, these improvements result in a specific yield for the 2T architecture in the amount of 2298 kWh/(kWp·a) for $E_g = 1.85$ eV being comparable with the case of the 4T architecture with a specific yield of 2279 kWh/(kWp·a) for $E_g = 1.8$ eV. Consequently, in a suitable bandgap range of 1.75 eV to 1.85 eV, the 2T architecture shows a comparable EY with its 4T competitor under realistic conditions, since the wide bandgap of the perovskite top absorber facilitates current matching significantly (see Figure S10). Results obtained for a perovskite bandgap of 1.8 eV (see Figure 7C) illustrate this finding for various locations. In some locations, the 2T architecture outperforms the 4T architecture by a few percent and in others vice versa. In general, the difference in EY provided by both architectures is vanishing. In view of maximizing the prospects of the technology, these results clearly underline the need for highly efficient wide bandgap perovskite absorber materials for the use in tandem configurations. At this point it is worth noting that yet such designed materials, as investigated in this study, do not exist. Wide bandgap mixed halide perovskites typically exhibit a FF < 80%³⁶ and suffer from a high deficit in open-circuit voltage due to photo-induced phase segregation.³⁷

Furthermore, our study reveals that the optimum perovskite bandgap for perovskite/CIGS tandem solar modules varies for realistic irradiation conditions in the range of 1.8 eV to 1.85 eV. For the 2T architecture, the optimum perovskite bandgap differs for various locations, eg, 1.85 eV in Phoenix (APE = 1.51 eV) and 1.8 eV in Portland (APE = 1.49 eV), since the APE of the investigated location affects the amount of current matching losses. Contrary to that, the 4T architecture does not display a dependence on the investigated location. In addition, the optimum perovskite bandgap is moderately higher for the 2T architecture, due to its dependence on current matching, than for the 4T architecture. When limiting the perovskite absorber thickness to more common thicknesses of 500 nm, the same optimum bandgap of 1.8 eV is identified (see Figure S9). These aspects accentuate the fact that the 4T architecture is more robust with regard to variations in spectral irradiation, whereas the 2T architecture promises a comparable EY if optimized appropriately.

4 | CONCLUSION

In this work, we present EY modelling of perovskite/CIGS tandem solar modules for various locations covering a broad range of climate zones.

Compared with reference CIGS SJ solar modules, $\text{CH}_3\text{NH}_3\text{PbI}_3$ /CIGS tandem solar modules with parameters taken from real state-of-the-art devices show a remarkable relative increase in EY of up to 41%, which highlights the great promises of this all thin-film technology. Comparing the 2T and 4T architecture, the former features a higher PCE under STC but a lower EY under realistic conditions. Our study identifies the spectral variations in realistic irradiation conditions as the main cause for the lower EY of the 2T architecture, despite featuring lower reflection and parasitic absorption losses compared with the 4T architecture. As a consequence, even the EY of optimized 2T device architectures is lower by 3.5% in average compared with the 4T architecture throughout all investigated locations.

One route to minimize the dependence of the 2T architecture on current matching and to enhance the overall performance of the technology is to optimize the bandgap of the perovskite top solar cell. In fact, our study, based on artificial wide-bandgap perovskite materials, shows a comparable EY for the 2T architecture with optimized perovskite absorber layers in the bandgap range of 1.75 eV to 1.85 eV to the 4T architecture. The relative increase in EY for bandgaps around 1.8 eV can be improved further to remarkable values of up to 52%. These findings highlight the importance of recent research on developing high efficiency wide bandgap perovskites for tandem PV.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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